Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

Aim :- Support Vector Machines (SVM)

1. **Implement the SVM algorithm for binary classification.**
2. **Train an SVM model using a given dataset and optimize its parameters.**
3. **Evaluate the performance of the SVM model on test data and analyze the results.**

Description :-

Step	Code / Function	Description
1	**pd.read_csv()**	Loads the Diabetes Prediction dataset.
2	**drop_duplicates()**	Removes duplicate records from the dataset.
3	**dropna()**	Removes records containing missing values.
4	**pd.get_dummies()**	Converts categorical features into numerical values.
5	**sample()**	Selects 5,000 records to reduce SVM memory usage.
6	**X and y**	Separates input features and the target `diabetes`.
7	**train_test_split()**	Divides the dataset into 80% training and 20% testing data.

Name :- Vinod Manohar Mali
Roll No :- T092

8	**StandardScaler()**	Standardizes the features before SVM training.
9	**SVC()**	Creates the Support Vector Machine classification model.
10	**GridSearchCV()**	Tests different SVM parameters to find the best model.
11	**grid_search.fit()**	Trains and optimizes the SVM model using training data.
12	**best_params_**	Displays the best SVM parameters.
13	**predict()**	Predicts diabetes classes for the test data.
14	**accuracy_score()**	Calculates the model's prediction accuracy.
15	**classification_report()**	Displays precision, recall, and F1-score.
16	`confusion_matrix()`	Shows correct and incorrect predictions.
17	`plt.imshow()`	Visualizes the confusion matrix.
18	`plt.show()`	Displays the final graph.

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

Code :-

```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
df = pd.read_csv("diabetes_prediction_dataset.csv")

print("Dataset loaded successfully!")

print("\nOriginal Dataset Shape:")
print(df.shape)
df = df.drop_duplicates()

print("\nDataset Shape After Removing Duplicates:")
print(df.shape)


df = df.dropna()

print("\nMissing values removed successfully!")
df = pd.get_dummies(
    df,
    columns=["gender", "smoking_history"],
    drop_first=True,
    dtype=int
```

```python
)

print("\nCategorical data converted successfully!")


if len(df) > 5000:
    df = df.sample(
        n=5000,
        random_state=42
    )

print("\nDataset used for SVM:")
print(df.shape)


X = df.drop(
    "diabetes",
    axis=1
)

y = df["diabetes"]


print("\nNumber of Features:")
print(X.shape[1])

print("\nTarget Classes:")
print(y.value_counts())
```

Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

```python
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)


print("\nTraining Samples:")
print(len(X_train))

print("\nTesting Samples:")
print(len(X_test))


pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("svm", SVC())
])


parameters = {
    "svm__C": [1, 10],
    "svm__kernel": ["linear", "rbf"]
}


grid_search = GridSearchCV(
```

```python
    pipeline,
    parameters,
    cv=3,
    scoring="accuracy",
    n_jobs=1
)


print("\n----------------------------------------")
print("Optimizing SVM Parameters...")
print("Please wait...")
print("----------------------------------------")


grid_search.fit(
    X_train,
    y_train
)


print("\nBest Parameters:")

print(
    grid_search.best_params_
)


print("\nBest Cross-Validation Accuracy:")

print(
```


Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

```
    round(
        grid_search.best_score_ * 100,
        2
    ),
    "%"
)



best_model = grid_search.best_estimator_


y_pred = best_model.predict(
    X_test
)


print("\nPrediction completed successfully!")


accuracy = accuracy_score(
    y_test,
    y_pred
)


print("\nTest Accuracy:")

print(
    round(
```

Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

```python
        accuracy * 100,
        2
    ),
    "%"
)


print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=[
            "No Diabetes",
            "Diabetes"
        ],
        zero_division=0
    )
)


cm = confusion_matrix(
    y_test,
    y_pred
)


print("\nConfusion Matrix:")
```

```python
print(cm)


plt.figure(
    figsize=(7, 6)
)

plt.imshow(cm)

plt.title(
    "SVM Confusion Matrix"
)

plt.xlabel(
    "Predicted Class"
)

plt.ylabel(
    "Actual Class"
)

plt.xticks(
    [0, 1],
    ["No Diabetes", "Diabetes"]
)

plt.yticks(
    [0, 1],
    ["No Diabetes", "Diabetes"]
```


Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College of Science

Subject :- Artificial Intelligence

Practical no 5

```
for i in range(2):

    for j in range(2):

        plt.text(
            j,
            i,
            str(cm[i, j]),
            ha="center",
            va="center"
        )


plt.colorbar()

plt.tight_layout()

plt.show()


print("\n========================================")
print("      SVM PRACTICAL COMPLETED")
print("========================================")

print(
    "Final Accuracy:",
    round(
        accuracy * 100,
        2
    ),
```

```
print(
    "Best Parameters:",
    grid_search.best_params_
)

print("========================================")
```

Name :- Vinod Manohar Mali

Roll No :- T092

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

Output :-

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 practical no 5.py ========
Dataset loaded successfully!

Original Dataset Shape:
(100000, 9)

Dataset Shape After Removing Duplicates:
(96146, 9)

Missing values removed successfully!

Categorical data converted successfully!

Dataset used for SVM:
(5000, 14)

Number of Features:
13

Target Classes:
diabetes
0    4567
1     433
Name: count, dtype: int64

Training Samples:
4000
```

```
Testing Samples:
1000

------------------------------------------
Optimizing SVM Parameters...
Please wait...
------------------------------------------

Best Parameters:
{'svm__C': 10, 'svm__kernel': 'linear'}

Best Cross-Validation Accuracy:
95.85 %

Prediction completed successfully!

Test Accuracy:
96.0 %

Classification Report:
              precision    recall  f1-score   support

 No Diabetes       0.96      1.00      0.98       913
    Diabetes       0.94      0.57      0.71        87

    accuracy                           0.96      1000
   macro avg       0.95      0.79      0.85      1000
weighted avg       0.96      0.96      0.96      1000
```

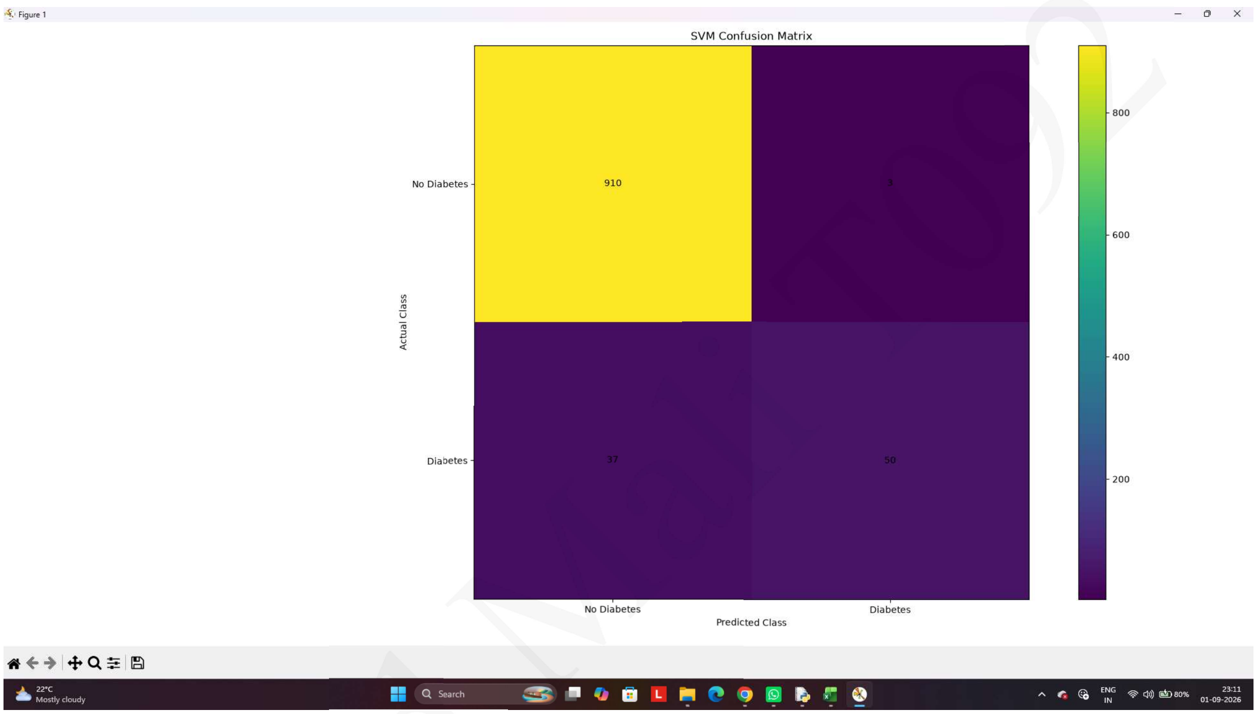
Name :- Vinod Manohar Mali

Roll No :- T092

Sheth L.U.J & Sir M.V College of Science
Subject :- Artificial Intelligence
Practical no 5

```
Confusion Matrix:
[[910   3]
 [ 37  50]]
```

Graph :-



Observation :-

- The confusion matrix shows that most cases are correctly classified.
- The diagonal values represent the correct predictions.
- The off-diagonal values represent incorrect predictions.
- Overall, the SVM model shows good classification performance on the test data.

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